

Design and Simulation of a Solar PV-Based Electric Vehicle Charging Station Using P&O MPPT and DC-DC Split- π Converter for Improved Battery Health

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Abstract—This paper deals with the design and simulation, via MATLAB, of a solar Photovoltaic (PV) based Electric Vehicle (EV) vehicle charging station with maximum power point tracking (MPPT) using the Perturb and Observe (P&O) algorithm. For improved battery health of the electric vehicle, a DC-DC split- π converter is used, with battery health quantified using Current Ripple Factor (CRF), Estimated Cycle Life (ECL) and State of Health (SoH). Three scenarios with varying irradiance and temperature values are used to test the robustness of the system under real-world conditions. Results show that CRF has reduced from more than 15% to less than 2%, with ECL improving from only about 150 cycles to more than 1700 cycles. This is also relates to SoH of around 90%, boosted from less than 7%. The MPPT efficiency has reached 78%, with entire system accomplishing an efficiency of 5.7%.

Index Terms—Solar PV, EV, P&O MPPT, DC-DC Converter, Split- π topology, ECL, CRF, SoH, CC-CV, Battery Health

I. INTRODUCTION

Electric Vehicles (EVs) are the backbone of the journey to electrification of the twenty first century, with global adoption at both professional and person. The reason for the this popularity of EVs lies mainly in the low running and maintenance costs, improved driving experiences, with owners benefiting from the convenience of charging at home [1]. This demands an infrastructure for charging that enables sustainability and efficiency, while maintaining excellent health of the battery.

To usher in aspects of high-efficiency and bi-directional flow, the DC-DC split- π converter topology is integrated into the proposed system. It is capable of both bucking (stepping-down) an boosting (stepping-up) voltages making it ideal for incorporation with EV batteries to smoothen both the charging and discharging cycles. The use of passive elements in a non-isolated system and operation at high-frequencies makes it inclined to be used in high-power density applications [2]. To quantify the improvement in battery health, parameters such as Current Ripple Factor (CRF), Estimated Cycle Life (ECL) and State of Health (SoH) are implemented.

Renewable energy systems underscore this sustainability factor by paving the way for the eradication of emission of Greenhouse Gases (GHGs), alongside enhanced reliability and stability. One such system is solar PV which converts sunlight into electricity using certain materials (for example, semiconductors like Silicon), through a phenomenon known as the Photovoltaic effect. As of October 2025, the solar capacity of India stands at 129 GW, accounting for more than half of the nation's total installed power capacity, marking the country's prowess in guaranteeing the shift to renewable energy. India also aims to install non-fossil fuel-based electricity capacity of 500 GW by 2050 [3].

However, the intermittent nature of these renewable energy sources, lead to unpredictable power output at the loads, with maximum values not attainable at all time points. Weather dependence, seasonal changes and day/night cycles all contribute to this intermittency. This constitutes major challenge especially for stability of the grid, as production does not always equal the requirements of the users.

To account for the intermittency of renewable energy and to ensure grid stability, Maximum Power Point Tracking (MPPT) Algorithm is employed in this paper. MPPT is an algorithm used in solar PV systems to extract the maximum power at all times, which works by continuously varying voltage and current (the electrical operating points) of the system to match the prevailing irradiance and temperature conditions. There are multiple MPPT techniques, with commonly used ones including Perturb and Observe (P&O) and Incremental Conductance. In this paper P&O is utilized, which periodically disturbs (perturbs) the operating voltage so that it can observe the resulting power change to track the maximum power point. It is low-cost and offers simple implementation, but presents minimized efficiency at low-irradiance and fast-charging conditions [4].

This paper combines the DC-DC split- π converter topology and the P&O MPPT algorithm in within a single sim-

ulation framework using MATLAB/Simulink R2022b with Simscape Electrical.

II. LITERATURE REVIEW

EVs play a two-way energy exchange role in its consideration with respect to smart grids. However, this integration creates some effects in the network, especially power-quality related effects. Power system stability is referred to the return of original, stable operating state of a power system after some kind of fault [5]. EVs introduce some non-linearity due to its load characteristics, leading to stability issues. Load behavior is also unpredictable, owing to charging times, power rating etc. These power-quality disturbances lead to high-frequency current ripple and interference in the CC-CV charging regime, which accelerate EV battery degradation, thereby reducing its cycle life. Narayan and Das [6] proposed a battery equalization and power transfer technique using a bi-directional DC-DC converter. This design not only improved thermal performance, but was also able to prevent State of Charge (SoC) imbalances to optimize battery life. Current ripple was shown to affect battery aging as investigated by De Breucker et. al [7], with the converter design goal is to reduce CRF, so that Estimated Cycle Life can be enhanced.

The use of split- π converter topology has been implemented extensively in literature due to its robustness in switching to buck, boost and buck-boost modes, without being prone to any kind of voltage and current fluctuations. It is prevalent in the applications of renewable systems, electric mobility and energy storage systems, such as in the interface of batteries and super-capacitors [8]. It also boasts a high conversion efficiency, and the ability to be connected in the parallel configuration, hence finding operating in multi-phase systems. The circuit consists of a buck topology, followed by a boost topology, with only two of the four switches conducting at an point of time, with the circuit topology depicted in Fig. 1. :

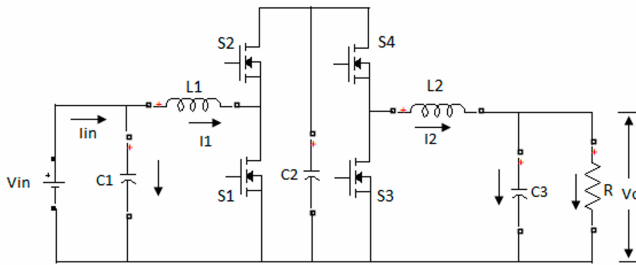


Fig. 1. Split- π Converter Circuit Topology [2]

It has three switching modes: buck, boost, and a pass-through mode which is detailed as follows [2]:

- **Boost Mode:** Switch S_4 is always ON, S_3 is always OFF and S_1 and S_2 are switched alternatively so that the boost voltage is always available at the output.
- **Buck Mode:** Switch S_2 is always ON, S_1 is always OFF and S_3 and S_4 are switched alternatively so that the buck voltage is always available at the output.

- **Pass-through Mode:** Switch S_1 and S_3 are permanently switched OFF and S_2 and S_4 are permanently switched ON, and the input voltage is directly passed to the output.

Monteiro et. al. proposed a Multi-level Di-directional and Symmetrical (MBS) split- π converter topology which can be operated using controlled variables of voltage and current with double of the switching frequency, validated by its performance considering all three switching modes [8].

Solar PV also contribute to non-linear behavior due to contingency on solar irradiance and ambient temperature. Thus, the extraction of utilization of output power from systems using PV cells also creates complications. Techniques such as MPPT were discovered to mitigate this issue, wherein maximum power can be continuously extracted with high efficiency. Menon, Prasad and Singh [9] analyzed the performance of an MPPT-integrated solar EV battery charger by comparing it to a standard charger without MPPT. The technique used was P&O, and the parameters evaluated was battery voltage, battery current, battery SoC and charge time. The P&O MPPT algorithm flow-chart can be visualized as follows:

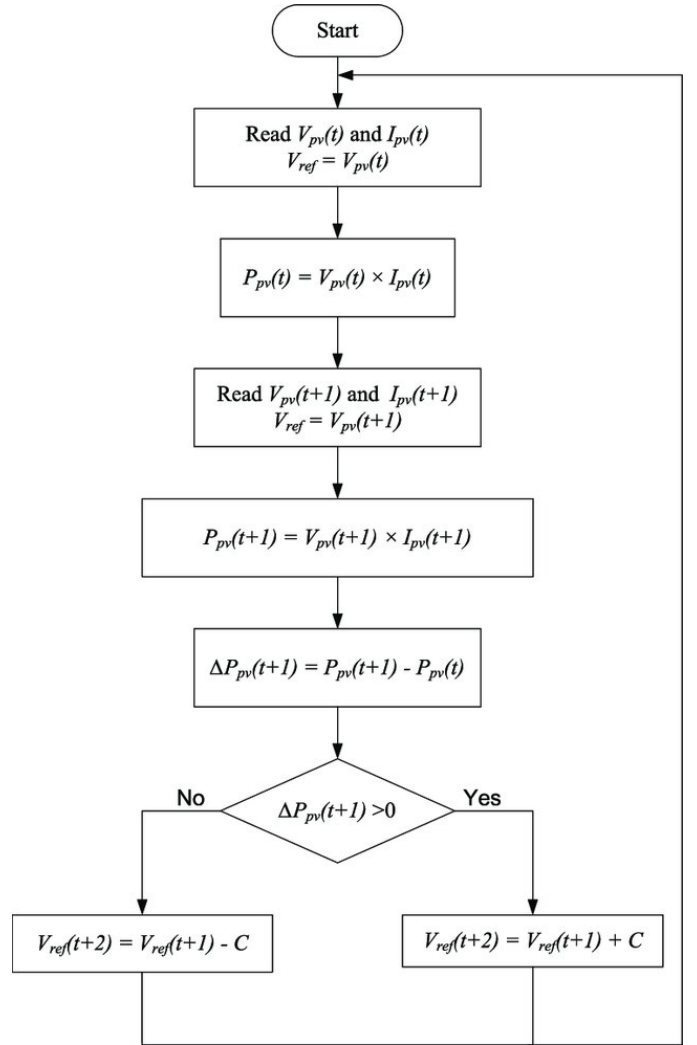


Fig. 2. The Flow-chart of P&O MPPT Algorithm [10]

III. METHODOLOGY

A. System Model and Architecture

The overall charging system comprises four major functional blocks as given below:

- PV Array Model
- P&O MPPT Controller
- DC-DC Split- π Converter
- Battery Charging Interface

The block diagram of the proposed system is as shown in Fig. 3.

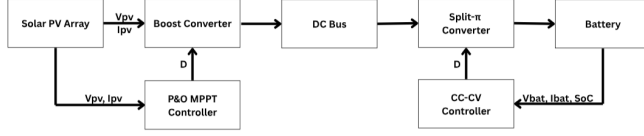


Fig. 3. Block Diagram of the Proposed System

B. Simulation Environment and Tools

- MATLAB/Simulink: Main simulation environment, Ver. R2022b
- Simscape Electrical: Specialised Power Systems
- Full Simulation Horizon: 3 Hours

C. Mathematical Formulation

1) PV Array Model

The PV cell output current is given by the standard single-diode model, as shown in Fig. 4.:

$$I_{pv} = I_L - I_0 \left(e^{\frac{V_{pv} + I_{pv} R_s}{n V_T}} - 1 \right) - \frac{V_{pv} + I_{pv} R_s}{R_{sh}} \quad (1)$$

where $I_L = 7.865$ A (light-generated current), $I_0 = 2.93 \times 10^{-10}$ A (saturation current), $n = 0.981$ (diode ideality factor), $V_t = 25$ mV (thermal voltage), $R_s = 0.394$ Ω (series resistance), and $R_{sh} = 313$ Ω (shunt resistance).

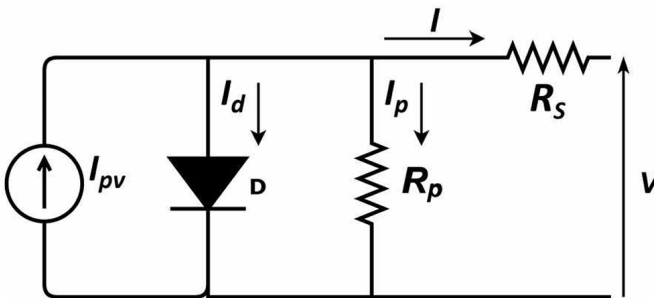


Fig. 4. Mathematical Model of Single diode Solar PV Cell [9]

These are the datasheet parameters for the Canadian Solar CS5P-220M module at STC (1000 W/m², 25°C).

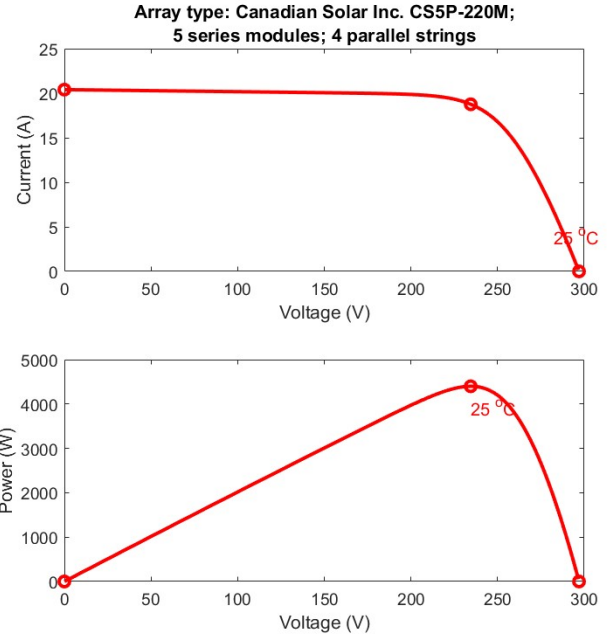


Fig. 5. I-V and P-V graphs of the Canadian Solar CS5P-220M module at STC (1000 W/m², 25°C)

2) Varying Conditions

The following three test scenarios define the varying conditions evaluated in this simulation, as depicted in Tabel I:

TABLE I
TEST SCENARIOS FOR PV SYSTEM

Case No.	Irradiance (W/m ²)	Temperature (°C)	Physical Scenario
1	1000	25	Baseline
2	1000 → 600	25	Sudden Cloud Cover
3	500 → 1000 → 800	25 → 45	Morning to Noon to Afternoon

3) P&O MPPT

The instantaneous PV power is given by:

$$P_{pv}(k) = V_{pv}(k) I_{pv}(k) \quad (2)$$

The incremental changes in power and voltage are defined as:

$$\Delta P = P_{pv}(k) - P_{pv}(k-1) \quad (3)$$

$$\Delta V = V_{pv}(k) - V_{pv}(k-1) \quad (4)$$

To reduce oscillations around the MPP and improve stability, the MPPT update is executed periodically:

$$\text{Update condition: } \text{mod}(k, N) = 0 \quad (5)$$

where $N = 30$ represents the update interval.

Additionally, a hysteresis condition is introduced to avoid unnecessary perturbations due to noise:

$$|\Delta P| > \epsilon \quad (6)$$

where $\epsilon = 0.5$ W.

To ensure safe operation of the converter, the duty cycle is constrained within practical limits:

$$D = \max(D_{\min}, \min(D_{\max}, D)) \quad (7)$$

where $D_{\min} = 0.05$ and $D_{\max} = 0.82$.

The relationship between the PV voltage and output voltage of the boost converter is approximated as:

$$V_{pv} = V_{out}(1 - D) \quad (8)$$

To maintain realistic operating bounds:

$$V_{pv} = \max(V_{\min}, \min(V_{oc}, V_{pv})) \quad (9)$$

where $V_{\min} = 50$ V and V_{oc} is the open-circuit voltage of the PV module.

4) Split- π Converter

The Split- π converter consists of L_1 , C_1 , L_2 , C_2 and four switches S_1 , S_2 , S_3 and S_4 operating at a switching frequency $f_s = 50$ kHz.

Component values are designed for $\Delta I_L < 10\%$ and $\Delta V_C < 5\%$.

5) Quantification of Battery Degradation

Battery degradation is quantified using three numerically measurable metrics:

- Current Ripple Factor (CRF):

$$CRF = \frac{\Delta I_{\text{chg,rms}}}{I_{\text{chg,avg}}} \times 100\% \quad (10)$$

Target: $CRF < 2\%$ with Split- π , compared to $CRF > 15\%$ without it.

- Estimated Cycle Life (ECL):

$$ECL = \frac{2000}{1 + 0.05 \cdot CRF^2} \quad (11)$$

Target: $ECL > 1700$ cycles with Split- π , compared to $ECL < 165$ without it.

- State of Health (SoH):

$$SoH = \frac{ECL}{2000} \times 100\% \quad (12)$$

Target: $SoH > 85\%$ cycles with Split- π , compared to $SoH < 8\%$ without it.

D. CC-CV Charging Algorithm and Duration Selection

The EV battery charging is implemented by the general CC-CV strategy. It is controlled to exhibit two modes, along with transition and termination conditions:

- **CC Mode:** The battery is charged at a constant current of $I_{\text{chg}} = 7$ A using a PI controller.
- **Transition Condition:** When the battery terminal voltage reaches $V_{\text{bat}} = 54$ V, the control transitions from CC mode to CV mode.
- **CV Mode:** In this phase, the battery voltage is maintained at $V_{\text{bat}} = 54$ V using a PI controller.

- **Termination Condition:** Charging is terminated when the battery current falls below $I_{\text{chg}} < 0.5$ A (C/40 rate).

Charge Modes are displayed by CC, CV and Done modes to indicate the battery charging during the full simulation horizon.

The proposed system is simulated for a duration of 3 hours, with the inclusion of a 1 second time-step. These values are identified on the basis of the following considerations:

- The 48 V, 20 Ah Li-ion battery, that was used in the proposed system, will require 1.5 to 2 hours to transition from 20% to 100% SoC, when charged at 7 A. This includes both CC and CV phases.
- If a shorter duration (for example, 1 hour) was applied, the complete CC-CV charging cycle would not be captured.
- Time-step of 1 second is added to avoid any kind of excessive computational complexity due to switching-level simulations.

E. Performance Metrics

The performance metrics, against which the proposed system is assessed, is depicted in Table II. The CRF, ECL and SoH is analyzed for two cases: with Split- π and without Split- π , for all three test scenarios of varying conditions.

TABLE II
PERFORMANCE METRICS

S.No	Metric	Definition
1	MPPT Efficiency	$\frac{P_{pv, \text{actual}}}{P_{pv, \text{max}}} \times 100\%$
2	Current Ripple Factor (CRF)	$\frac{\Delta I_{\text{chg,rms}}}{I_{\text{chg,avg}}} \times 100\%$
3	Estimated Cycle Life (ECL)	$\frac{2000}{1 + 0.05 \cdot CRF^2}$
4	State of Health (SoH)	$\frac{ECL}{2000} \times 100\%$
5	System Efficiency	$\frac{P_{\text{battery}}}{P_{pv, \text{max}}} \times 100\%$

IV. RESULTS AND DISCUSSIONS

A. Overview

The simulation was run in MATLAB, with the Simulink model shown in Fig.6., and the following plots were derived:

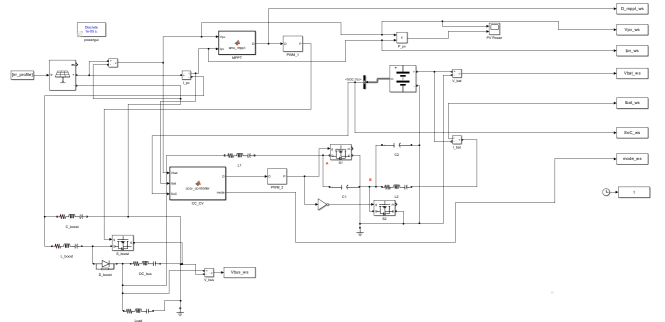


Fig. 6. Complete Simulink Model of the Proposed System

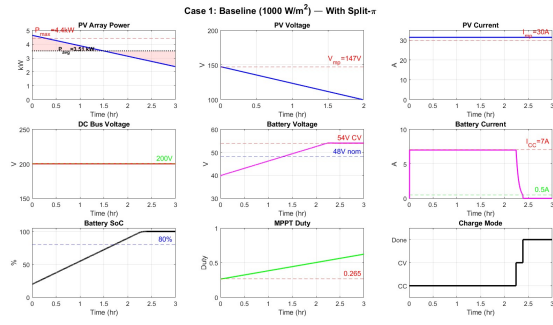


Fig. 7. Plots for Case I (1000 W/m^2) with Split- π , including PV Array Power (with losses highlighted), PV Voltage and Current, DC Bus Voltage, Battery Voltage, Current and SoC, MPPT Duty and Charge Mode

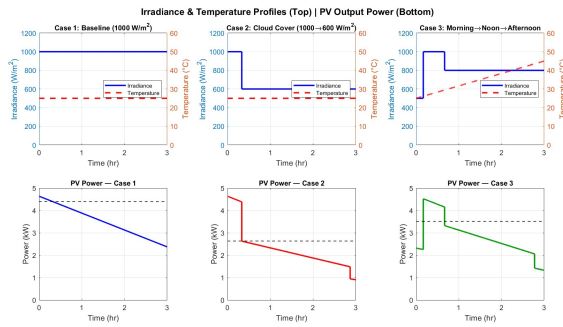


Fig. 8. Plots showing Irradiance and Temperature profiles of all three cases and corresponding PV Output Power

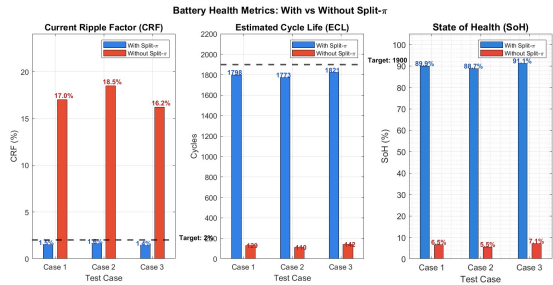


Fig. 9. Comparison of Battery Health Metrics (CRF, ECL and SoH) with and without Split- π

Under all operating conditions, the resultant PV array power tracks the theoretical maximum power, establishing the efficiency of the P&O MPPT algorithm. As shown in Fig. 7. demonstrating results of Case I, an average power of 3.5 kW is extracted, which is nearly equal to the maximum power of 4.4 kW. The shaded red region shows the power losses associated with the system.

The PV voltage and current remain borderline to theoretical values of ($V_{pv} = 147 \text{ V}$ and $I_{pv} \approx 30 \text{ A}$), but deviations are present due to the introduction of small oscillations around the maximum power point, which is on account of the P&O algorithm.

Under varying irradiance conditions, the system demonstrates robustness in its performance, with sudden rise and drop in irradiance showing a proportional rise and drop in PV output power, as shown in Fig. 8. The PV output power is shown to decrease slightly even though irradiance is constant. DC bus voltage slightly increases above 200V due to the CC-CV controller, which surges its duty cycle to push current into the rising battery voltage. The MPPT then counters this by raising its duty cycle, which reduces the V_{pv} value, thereby leading to the operating point drifting away from the maximum power point.

The DC bus voltage is maintained at the reference value of 200 V throughout all the three scenarios.

The battery charging profile follows the general CC-CV behavior. During the CC phase, the charging current is sustained at a constant of 7 A, with increasing voltage. Once the threshold voltage has reached a value of 54 V, the charging transitions to the CV phase, leading to decrease in current. When current falls below the threshold of 0.5 A, charging is stopped, and the plot shows DONE, indicating full charge. The SoC shows a linear increase from 20% to 100%.

B. Comparative Analysis of Test Scenarios

1) Case I: Baseline:

At an irradiance of 1000 W/m^2 and a temperature of 25°C , these are the derived outputs:

- MPPT Efficiency: 78.44%
- System Efficiency: 5.70%
- CRF (With Split- π): 1.50%
- ECL (With Split- π): 1798
- SoH (With Split- π): 89.9%
- CRF (Without Split- π): 17.00%
- ECL (Without Split- π): 129
- SoH (Without Split- π): 6.5%

2) Case II: Sudden Cloud Cover:

At an irradiance of 1000 W/m^2 dropped to 600 W/m^2 (due to sudden cloud cover) and a temperature of 25°C , these are the derived outputs:

- MPPT Efficiency: 77.50%
- System Efficiency: 5.70%
- CRF (With Split- π): 1.60%
- ECL (With Split- π): 1773
- SoH (With Split- π): 88.7%
- CRF (Without Split- π): 18.50%
- ECL (Without Split- π): 110
- SoH (Without Split- π): 5.5%

3) Case III: Morning to Noon to Afternoon:

At an irradiance profile of 500 W/m^2 in the morning, raised to 1000 W/m^2 at noon and dropped to 800 W/m^2 in the afternoon, and an ascending temperature profile from 25°C to 45°C , these are the derived outputs:

- MPPT Efficiency: 77.64%
- System Efficiency: 5.70%

- CRF (With Split- π): 1.40%
- ECL (With Split- π): 1821
- SoH (With Split- π): 91.1%
- CRF (Without Split- π): 16.20%
- ECL (Without Split- π): 142
- SoH (Without Split- π): 7.1%

C. Discussion of Results

On observation of the proposed system and the three test scenarios, it is justifiable that the P&O MPPT effectively tracks the maximum power points with efficiencies in the range of 77-78%. A slight deviation from ideal performance occurs due to the oscillatory behavior of the P&O algorithm.

The Split- π converter topology significantly improved the battery health metrics with all three cases having CRF values less than 2%, leading to ECL values about 1700. This in turn is exhibited in SoH values around 90%. The ECL formula is a curve-fit to that empirical dataset and it is not a general model, therefore the derived results is disproportionate to real-life circumstances.

However, the system has a very low efficiency primarily due to the mismatch between PV power available and the battery charging capacity.

D. Limitations of the Proposed System

Although the proposed system is effective with positive outcomes, there are some limitations that exist in the present work. These can be listed as follows:

- The converter is modeled using an averaged approach with no high-frequency switching dynamics.
- Thermal effects on battery degradation are not considered.
- The battery is represented using a simplified equivalent circuit model with constant parameters.
- Converter losses such as switching losses and conduction losses are not included.
- Measurement noise and control delays are neglected.

Excluding these complexities enables the simulation to output efficient results, but may produce deviations from real-life performances.

E. Future Works

Any kind of future work on the proposed system is to be done to improve its real-life applicability. Incorporating high-frequency effects can be done to enhance the accuracy of the system.

Thermal modeling of the battery can be integrated so that study of any kind of temperature-dependent degradation mechanisms can be done.

MPPT technique can be changed, wherein P&O can be replaced with Incremental Conductance, or any Artificial Intelligence (AI) based approaches.

A closed-loop bus voltage regulator can be introduced (in hardware implementation) to eliminate the duty drift of MPPT.

Experimental validation can be done using hardware prototypes to strengthen the practical applicability of the proposed model.

V. CONCLUSION

The paper titled "Design and simulation of a solar PV-based EV Charging System using P&O MPPT and DC-DC Split- π Converter for Improved Battery Health" evaluated the proposed system under multiple operating conditions of varying irradiance and temperature profiles.

The results demonstrate that the MPPT algorithm effectively tracks the maximum power point under all three test scenarios, with the battery charging process follows the general CC-CV profile.

A significant part of the proposed system is the quantification of battery health using CRF and ECL. The Split- π converter reduces current ripple to below 2%, resulting in a considerable improvement in estimated cycle life beyond 1700 cycles and overall boosted SoH of 90%.

Although system efficiency appears limited due to power mismatch between the PV array and battery charging capacity, the overall proposed system demonstrates robust performance, improved battery health, and adaptability under varying environmental conditions.

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